



737-800
STRUCTURAL REPAIR MANUAL
ALLOWABLE DAMAGE 1 - FUSELAGE SKIN

1. Applicability

- A. Allowable Damage 1 gives the allowable damage and the airplane flight operation limits for the fuselage skins shown in Fuselage Skin, Figure 101/ALLOWABLE DAMAGE 1.
- B. Allowable Damage 1 is not applicable to the skin locations that follow:
 - (1) In Section 43 between Stations 360-540 and between Stringers 24L-24R more than 3.0 in. (76.0 mm) from a door cutout.
 - (2) In Section 46 between Stations 727-887 and between Stringers 23L-23R more than 3.0 in. (76.0 mm) from a door cutout.
 - (3) For airplanes that have a flat aft pressure bulkhead, in Section 48 between Stations 1138-1156 and between Stringers 4R-5R.

2. General

- A. Refer to Allowable Damage Zones, Figure 102/ALLOWABLE DAMAGE 1 for the allowable damage zones.
- B. Refer to Paragraph 4./ALLOWABLE DAMAGE 1 for the allowable damage limits.
- C. If you find damage, airplane flight operation limits can be necessary. Refer to Paragraph 5./ALLOWABLE DAMAGE 1
- D. Make sure the aerodynamic smoothness is satisfactory or there will be a decrease in the performance of the airplane. Refer to 51-10-01.

NOTE: The areas specified in Paragraph 4./ALLOWABLE DAMAGE 1 will be in a critical or non-critical aerodynamic area. Refer to Allowable Damage Zones, Figure 102/ALLOWABLE DAMAGE 1 for the different areas.

- E. After you remove the damage on aluminum parts, do the steps that follow:
 - (1) Apply a chemical conversion coating to the reworked areas. Refer to 51-20-01.
 - (2) Apply two layers of BMS 10-79, Type III primer to the reworked areas. Refer to SOPM 20-44-04.
 - (3) Apply BMS 5-95 sealant to the mating surfaces before you assemble a lap joint. Refer to 51-20-05.
- F. Fill the space between the edges of skin panels in the area forward of BS 663.75 with BMS 5-95 sealant. Refer to 51-20-05.
- G. Refer to 51-30-03 for possible sources of the abrasive and other materials you can use to remove the damage.
- H. Refer to 51-30-05 for possible sources of the equipment and tools you can use to remove the damage.

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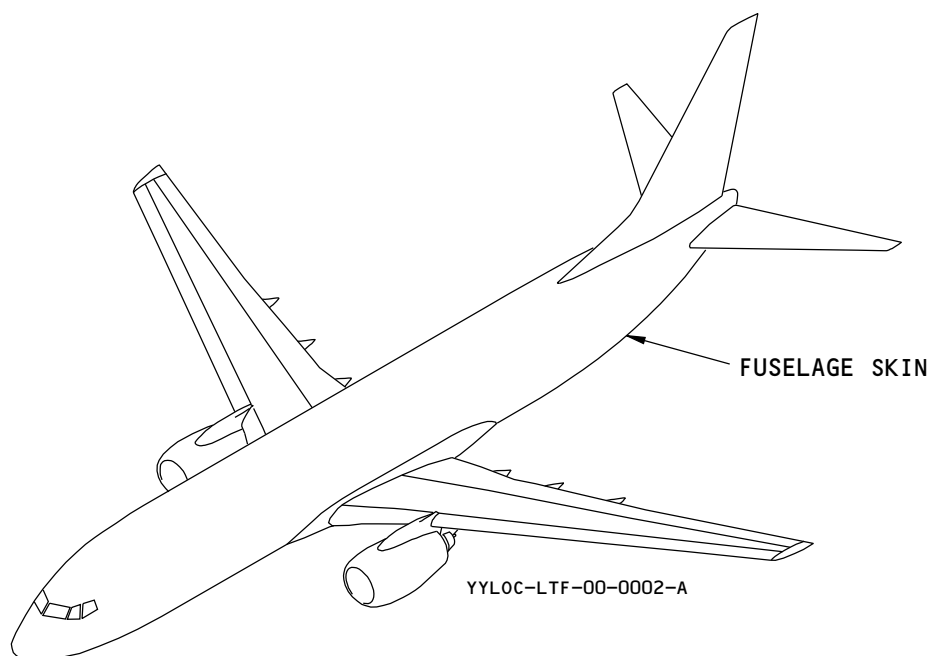
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**Fuselage Skin
Figure 101**

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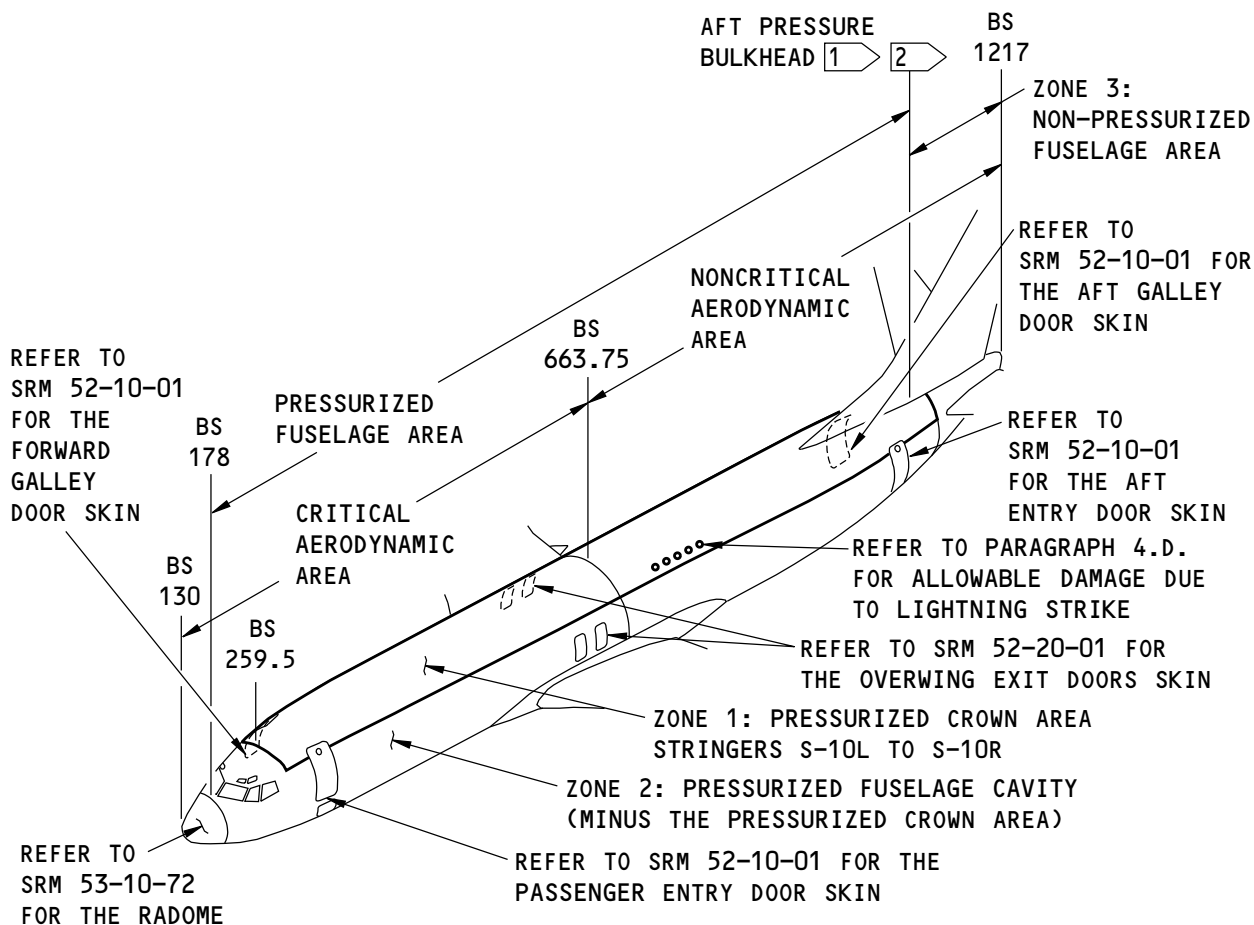
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Allowable Damage Zones
Figure 102

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3. References

Reference	Title
51-10-01	AERODYNAMIC SMOOTHNESS
51-10-01, GENERAL	Aerodynamic Smoothness Requirements
51-20-01	PROTECTIVE TREATMENT OF METALLIC AND COMPOSITE MATERIALS
51-20-05	REPAIR SEALING
51-30-03	NON-METALLIC MATERIALS
51-30-05	EQUIPMENT AND TOOLS FOR REPAIRS
51-40-05, GENERAL	Fastener Hole Sizes
51-70-01	REPAIRS FOR MINOR DENTS IN METALLIC SHEET MATERIALS
53-00-01, REPAIR 13	MOVED - Repair of Lightning Strike Damage in Fuselage Skin

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Reference	Title
AMM 51-21-99 P/B 701	DECORATIVE EXTERIOR PAINT SYSTEM - CLEANING/PAINTING
SOPM 20-44-04	Application of Urethane Compatible Primer

4. Allowable Damage Limits for the Fuselage Skin

NOTE: See Paragraph 4.D./ALLOWABLE DAMAGE 1 for allowable damage due to lightning strike to all zones of the fuselage skin and fasteners.

A. Zone 1 - (Pressurized Fuselage Crown Area)

(1) Cracks:

- (a) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A and B.

(2) Nicks, Gouges, Scratches, and Corrosion:

- (a) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A, B, C, D, E, F, T, U, and V.

1) You can remove the damage from the areas that follow:

- a) The inner and outer surfaces of the solid skin, a bonded skin assembly, or a skin lap splice
- b) The sealed surfaces of a skin lap splice.

- 2) The total depth of the damage removed at each location must be less than or equal to the percentage of the skin thickness at the damage location.

NOTE: When you calculate the depth of damage removal, use the skin thickness given in the applicable identification subject or engineering drawing.

- a) When you remove damage on a solid skin, use the skin thickness at the damage location to calculate the percentage of damage removal.
- b) When you remove damage on a bonded skin assembly, use the thickness of the skin layer from which you removed the damage to calculate the percentage of damage removal.

- (b) If you remove corrosion damage that goes across the head of a countersink rivet, replace the rivet.

- (c) If you remove a countersink rivet to remove the damage, install a new rivet of the same size and type as the initial rivets. Use an oversize rivet as necessary.

- 1) The countersink depth must not be more than 80 percent of the thickness of the solid skin.
- 2) The countersink depth must not be more than 80 percent of the total thickness of the bonded skin assembly.
- 3) You can replace the countersink rivet with a protruding head rivet. Refer to Table 101/ALLOWABLE DAMAGE 1 for the type and size of protruding head rivet to use.

NOTE: A mixture of 3/16 inch fasteners and 1/4 inch fasteners is not permitted in lap or butt splices. You are permitted to replace initial 3/16 inch fasteners with 7/32 inch oversize fasteners on lap and butt splices.

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Table 101:

REPAIR RIVETS	
INITIAL RIVET	REPAIR RIVET
BACR15GF5D	BACR15FT7D
BACR15GF6D	BACR15FT8D

(3) Dents:

- (a) Dents in extra critical areas near the air data sensors must be reworked to initial contour as described in 51-10-01, GENERAL paragraph 5.



CAUTION

DO NOT FILL DENTS THAT ARE MORE THAN THE ALLOWABLE DAMAGE LIMITS. IF YOU DO NOT OBEY, THE RESULT CAN BE AN UNSATISFACTORY CONDITION THAT CAN CAUSE MORE DAMAGE TO THE AIRPLANE STRUCTURE.

- (b) Other dents not near the air data sensors are permitted as given in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G and Table 102/ALLOWABLE DAMAGE 1.
- 1) A dent larger than the limits shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G is permitted if:
 - a) There are no loose or missing fasteners
 - b) There are no damaged fastener holes
 - c) There are no creases, gouges or cracks near the dent
 - d) You do not fill the dent
 - e) You make an inspection of the dent for corrosion and cracks at the flight cycle intervals as given in Table 102/ALLOWABLE DAMAGE 1.
 - f) The damage does not go across a stringer, frame, or intercostal

Table 102:

FUSELAGE DENT ALLOWABLE DAMAGE LIMITS ^{*[1]}		
DEPTH	W/Y RATIO	CORRECTIVE ACTION
Less than 0.125 in. (3.175 mm)	Greater than 30	Allowable Damage. No repair necessary.
Less than 0.125 in. (3.175 mm)	W/Y Ratio between 15 and 30 (15 < W/Y < 30)	Dent must be repaired permanently. However, you can delay the repair if you do all the steps that follow: 1. You initially do a Detailed Visual Inspection of the dent. 2. You do an HFEC inspection within 5,000 cycles. 3. You do a Detailed Visual Inspection every 4,000 cycles. 4. A permanent repair is made before 10,000 cycles.
Less than 0.125 in. (3.175 mm)	Less than 15	Dent must be repaired permanently. However, you can delay the repair if you do all the steps that follow: 1. You initially do a Detailed Visual Inspection of the dent. 2. You do an HFEC inspection within 5,000 cycles. 3. You do a Detailed Visual Inspection every 4,000 cycles. 4. A permanent repair is made before 10,000 cycles.
Depth between 0.125 in. (3.175 mm) and 0.25 in. (6.35 mm) (0.125 < Depth < 0.25)	W/Y Ratio between 15 and 30 (15 < W/Y < 30)	Dent must be repaired permanently. However, you can delay the repair if you do all the steps that follow: 1. You initially do a Detailed Visual Inspection of the dent. 2. You do an HFEC inspection within 5,000 cycles. 3. You do a Detailed Visual Inspection every 4,000 cycles. 4. A permanent repair is made before 10,000 cycles.
Depth between 0.125 in. (3.175 mm) and 0.25 in. (6.35 mm) (0.125 < Depth < 0.25)	Greater than 30	Dent must be repaired permanently. However, you can delay the repair if you do all the steps that follow: 1. You initially do a Detailed Visual Inspection of the dent. 2. You do an HFEC inspection within 5,000 cycles. 3. You do a Detailed Visual Inspection every 4,000 cycles. 4. A permanent repair is made before 13,000 cycles.

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Table 102: (Continued)

FUSELAGE DENT ALLOWABLE DAMAGE LIMITS ^{*[1]}		
DEPTH	W/Y RATIO	CORRECTIVE ACTION
Greater than 0.125 in. (3.175 mm)	Less than 15	Dent must be repaired permanently. However, you can delay the repair if you do all the steps that follow: 1. You initially do a Detailed Visual Inspection of the dent. 2. You do an HFEC inspection within 500 cycles. 3. You do a Detailed Visual Inspection every 500 cycles. 4. A permanent repair is made before 5,000 cycles.
Greater than 0.25 in. (6.35 mm)	All	

^{*[1]} In order to use Table 102/ALLOWABLE DAMAGE 1, dent must meet these requirements.

Dents must:

- Be smooth and free of sharp creases, gouges, or cracks.
- Be 10 in. (254 mm) or more from a cutout and 3 in. (76 mm) or more from a skin splice as measured from the nearest fastener hole centerline to the edge of the dent periphery.
- Not be common to a stringer, tear strap, frame, or fastener.

Dents must not have:

- Broken, pulled, or loose fasteners.
 - Elongated fastener holes.
 - Substructure damage.
- (4) Holes and Punctures are not permitted.
 - (5) Wrinkles and Buckles are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail H if the damage does not go in or through a fastener hole.
 - (6) Unbonded areas:
 - (a) Unbonded areas in the skin-to-bear strap bonds at the forward and aft entry and galley door surround structure are permitted.
 - 1) You must apply BMS 3-23 corrosion inhibiting compound to an unbonded area that is at an edge.
 - (b) Unbonded areas in skin-to-doubler bonds are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail I.
 - 1) You must apply BMS 3-23 corrosion inhibiting compound to an unbonded area that is at an edge.

B. Zone 2 - (Pressurized Fuselage not in the Crown Area)

(1) Cracks:

- (a) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A and B.
- (b) Remove the damage as shown in Figure 103, Details Q, R, and S for the fuselage skin areas adjacent to the forward and aft cargo doors. Make sure to remove the damage with a surface finish of 63 Ra or better. For Figure 103, Detail R, do a High Frequency Eddy Current (HFEC) inspection of the reworked area of the skin and the fastener holes that were removed to make sure there is no further damage. Refer to NDT Part 6, 51-00-00, Figure 16.

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- 1) Replace any fasteners in the damaged area with initial type drawing fasteners in a shifted transition fit hole. Use an interference between 0.0005 and 0.0045 inch. Refer to 51-40-05, GENERAL.

NOTE: Install a SRM approved hex drive bolt and nut to damaged areas along the aft edge of the door cutout as given in Figure 103, Detail Q that is common to a frame outer chord fastener. Torque as given in BAC5004-2 if the surface sealant has been applied again between the frame chord and the adjacent skin surfaces.

(2) Nick, Gouges, Scratches, and Corrosion:

- (a) Remove the damage as given in Figure 103, Details Q, R, and S for fuselage skin areas adjacent to the forward and aft cargo doors. Make sure to remove the damage with a surface finish of 63 Ra or better. For Figure 103, Detail R, do a High Frequency Eddy Current (HFEC) inspection of the reworked area of the skin and the fastener holes that were removed to make sure there is no further damage. Refer to NDT Part 6, 51-00-00, Figure 16.

- 1) Replace any fasteners in the damaged area with initial type drawing fasteners in a shifted transition fit hole. Use an interference between 0.0005 and 0.0045 inch. Refer to 51-40-05, GENERAL.

NOTE: Install a SRM approved hex drive bolt and nut to damaged areas along the aft edge of the door cutout as given in Figure 103, Detail Q that is common to a frame outer chord fastener. Torque as given in BAC5004-2 if the surface sealant has been applied again between the frame chord and the adjacent skin surfaces.

- (b) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A, B, C, D, E, T, U, and V.

- 1) You can remove the damage from the areas that follow:

- a) The inner and outer surfaces of the solid skin, a bonded skin assembly, or a skin lap splice.
- b) The sealed surfaces of a skin lap splice.

- 2) The total depth of the damage removed at each location must be less than or equal to the percentage of the skin thickness at the damage location.

NOTE: When you calculate the depth of damage removal, use the thickness given in the applicable identification subject or engineering drawing.

- a) When you remove damage on a solid skin, use the skin thickness at the damage location to calculate the percentage of damage removal.
- b) When you remove damage on a bonded skin assembly, use the thickness of the skin layer from which you removed the damage to calculate the percentage of damage removal.

- (c) If you remove corrosion damage that goes across the head of a countersink rivet, replace the rivet.
- (d) If you remove a countersink rivet to remove the damage, install a new rivet of the same size and type as the initial rivets. Use an oversize rivet as necessary.
 - 1) The countersink depth must not be more than 80 percent of the thickness of the solid skin.
 - 2) The countersink depth must not be more than 80 percent of the total thickness of a bonded skin assembly.

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- 3) You can replace the countersink rivet with a protruding head rivet. Refer to Table 101/ALLOWABLE DAMAGE 1 for the type and size of protruding head rivet to use.

NOTE: Protruding head fasteners are not permitted near static ports or angle of attack sensors where external repairs are not permitted. Refer to 51-10-01. A mixture of 3/16 inch fasteners and 1/4 inch fasteners is not permitted in lap or butt splices. You are permitted to replace initial 3/16 inch fasteners with 7/32 inch oversize fasteners on lap and butt splices.

(3) Dents:

- (a) Dents in extra critical areas near the air data sensors must be reworked to initial contour as described in 51-10-01, GENERAL paragraph 5.



CAUTION

DO NOT FILL DENTS THAT ARE MORE THAN THE ALLOWABLE DAMAGE LIMITS. IF YOU DO NOT OBEY, THE RESULT CAN BE AN UNSATISFACTORY CONDITION THAT CAN CAUSE MORE DAMAGE TO THE AIRPLANE STRUCTURE.

- (b) Other dents not near the air data sensors are permitted as given in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G and Table 102/ALLOWABLE DAMAGE 1.
- 1) A dent larger than the limits shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G is permitted if:
- a) There are no loose or missing fasteners
 - b) There are no damaged fastener holes
 - c) There are no creases, gouges, or cracks near the dent
 - d) The damage does not go across stringers, frames, or intercostals
 - e) You do not fill the dent.
 - f) You make an inspection of the dent for corrosion and cracks at the flight cycle intervals as given in Table 102/ALLOWABLE DAMAGE 1.
- (4) Wrinkles and Buckles are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail H if:
- (a) They do not go in or through a fastener hole
 - (b) They do not cross stringers, intercostals or frame
 - (c) There are no cracks near the wrinkles or buckles
 - (d) They are not more than 0.15 inch in depth as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail H
 - (e) There are no loose or damaged fasteners
 - (f) You make an inspection of the dent for corrosion and cracks at the flight cycle intervals that follow:
 - 1) For airplanes that have completed Service Bulletin 737-21-1149, inspect each 2500 flight cycle interval or more frequently.
 - 2) For airplanes that have not completed Service Bulletin 737-21-1149, inspect each 5000 flight cycle interval or more frequently.
- (5) Holes and Punctures are not permitted.
- (6) Unbonded areas:

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- (a) Unbonded areas in the skin-to-bear strap bonds at the forward and aft entry and galley door surround structure are permitted.
 - 1) You must apply BMS 3-23 corrosion inhibiting compound to an unbonded area that is at an edge.
 - (b) Unbonded areas in skin-to-doubler bonds are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail I .
 - 1) You must apply BMS 3-23 corrosion inhibiting compound to an unbonded area that is at an edge.
- C. Zone 3 - (Non-pressurized Fuselage)
- (1) Cracks:
 - (a) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A and B .
 - (2) Nicks, Gouges, Scratches, and Corrosion:
 - (a) Remove the damage as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Details A, B, C, D, E, T, U, and V.
 - 1) You can remove the damage from the areas that follow:
 - a) The inner and outer surfaces of the solid skin, a bonded skin assembly, or a skin lap splice
 - b) The sealed surfaces of a skin lap splice.
 - 2) The total depth of the damage removed at each location must be less than or equal to the percentage of the skin thickness at the damage location.

NOTE: When you calculate the depth of damage removal, use the skin thickness given in the applicable identification subject or engineering drawing.

 - a) When you remove damage on a solid skin, use the skin thickness at the damage location to calculate the percentage of damage removal.
 - b) When you remove damage on a bonded skin assembly, use the thickness of the skin layer from which you removed the damage to calculate the percentage of damage removal.
 - (b) If you remove a countersink rivet to remove the damage, install a new rivet of the same size and type as the initial rivets. Use an oversize rivet as necessary.
 - 1) The countersink depth must not be more than 80 percent of the thickness of the solid skin.
 - 2) The countersink depth must not be more than 80 percent of the total thickness of a bonded skin assembly.
 - 3) You can replace the countersink rivet with a protruding head rivet. Refer to Table 101/ALLOWABLE DAMAGE 1 for the type and size of protruding head rivet to use.

NOTE: A mixture of 3/16 inch fasteners and 1/4 inch fasteners is not permitted in lap or butt splices. You are permitted to replace initial 3/16 inch fasteners with 7/32 inch oversize fasteners on lap and butt splices.
 - (3) Dents
 - (a) Dents in extra critical areas near the air data sensors must be reworked to initial contour as described in 51-10-01, GENERAL paragraph 5.

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CAUTION

DO NOT FILL DENTS THAT ARE MORE THAN THE ALLOWABLE DAMAGE LIMITS. IF YOU DO NOT OBEY, THE RESULT CAN BE AN UNSATISFACTORY CONDITION THAT CAN CAUSE MORE DAMAGE TO THE AIRPLANE STRUCTURE.

- (b) Other dents not near the air data sensors are permitted as given in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G and Table 102/ALLOWABLE DAMAGE 1.
 - 1) A dent larger than the limits shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail G is permitted if:
 - a) There are no loose or missing fasteners
 - b) There are no damaged fastener holes
 - c) There are no creases, gouges, or cracks near the dent
 - d) You do not fill the dent
- (4) Holes and Punctures are permitted if:
 - (a) They are a maximum of 0.25 inch in diameter
 - (b) They are a minimum of 1.00 inch away from a fastener hole, an edge, or other damage
 - (c) They are filled with a 2017-T3 or 2017-T4 aluminum protruding head rivet installed without sealant.
- (5) Wrinkles or Buckles are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail H if:
 - (a) The damage does not go in or through a fastener hole
 - (b) You make an inspection of the damage for corrosion and cracks at each 5000 flight cycle interval or more frequently.
- (6) Unbonded areas in skin-to-doubler bonds are permitted as shown in Allowable Damage Limits, Figure 103/ALLOWABLE DAMAGE 1, Detail I.
 - (a) You must apply BMS 3-23 corrosion inhibiting compound to an unbonded area that is at an edge.

Table 103:

BLENDING LIMITS FOR FIGURE 104				
BODY SECTION	STRINGERS	SIDE OF THE FUSELAGE	BODY STATION	MAXIMUM DAMAGE DEPTH PERMITTED
43	S-17 TO S-18 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	L	STA 500E TO STA 500F	0.05T
43	S-21 TO S-22 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	L	STA 500F TO STA 500G	0.05T
43	S-13 TO S-15	BOTH	STA 500H TO STA 500I	0.05T
43	S-13 TO S-17	R	STA 520 TO STA 540	0.05T
43	S-14 TO S-18	L	STA 520 TO STA 540	0.05T

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Table 103: (Continued)

BLENDING LIMITS FOR FIGURE 104				
BODY SECTION	STRINGERS	SIDE OF THE FUSELAGE	BODY STATION	MAXIMUM DAMAGE DEPTH PERMITTED
46	S-2 TO S-4 (For Airplanes That Have Not Completed Service Bulletin 737-21-1149)	L	STA 847 TO STA 887	0.05T
46	S-1 TO S-4 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	BOTH	STA 847 TO STA 887	0.05T
46	S-3 TO S-4	L	STA 807 TO STA 847	0.05T
46	S-5 TO S-7 (For Airplanes That Have Not Completed Service Bulletin 737-21-1149)	L	STA 767 TO STA 807	0.05T
46	S-5 TO S-7 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	BOTH	STA 727I TO STA 807	0.05T
46	S-6 TO S-7	L	STA 867 TO STA 887	0.05T
46	S-8 TO S-9 (For Airplanes That Have Not Completed Service Bulletin 737-21-1149)	L	STA 727H TO STA 727I	0.05T
46	S-7 TO S-9 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	BOTH	STA 727H TO STA 727I	0.05T
46	S-8 TO S-9 (For Airplanes That Have Not Completed Service Bulletin 737-21-1149)	L	STA 727H TO STA 727J	0.05T
46	S-7 TO S-9 (For Airplanes That Have Completed Service Bulletin 737-21-1149)	R	STA 747 TO STA 767	0.05T
46	S-14 TO S-16	R	STA 827 TO STA 847	0.05T
46	S-17 TO S-19	BOTH	STA 727A TO STA 727B	0.05T
48 (For Airplanes That Have a Flat Aft Pressure Bulkhead)	S-4 to S-5	L	STA 1138 to STA 1156	0.05T

D. For allowable damage for heat damage due to lightning strike do as follows:

NOTE: These limits are applicable to all Zones of the aircraft.

(1) Do the following general inspections:

- Do a detail visual inspection of the damage with a minimum of 10X magnification in a 3 inch radius area around the damage to make sure there are no cracks.
- Do a general visual inspection of the skin within a 20 inch radius around the damage to make sure there are no cracks or corrosion.

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- (2) For lightning strike damage to the fuselage skin away from fasteners, you can operate the airplane for 350 flight cycles if you meet the following conditions:
 - (a) The lightning strike damage is equal to or less than 0.250 inch in diameter,
 - (b) The distance between any two lightning strike damages is greater than 3 inches, and
 - (c) The lightning strike damage is a minimum of 1 inch away from an edge of a part or fasteners.
 - (d) No skin material is missing.
- (3) For lightning strike damage to the fuselage at fasteners you can operate the airplane for up to 350 flight cycles if you meet the following conditions:
 - (a) The damage at a fastener is within the damage rating limits given in Figure 103, Details L and M with examples given on Details N, O, and P.
 - (b) You do a close visual inspection to the damage area to make sure there are no exposed countersink fasteners or further damage.
 - (c) You apply all initial production drawing finishes as necessary.
- (4) You must do the repair given in 53-00-01, REPAIR 13 at or before 350 flight cycles for all lightning strike damages.

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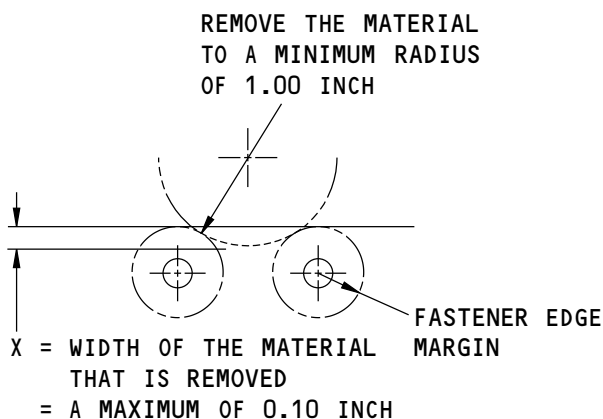
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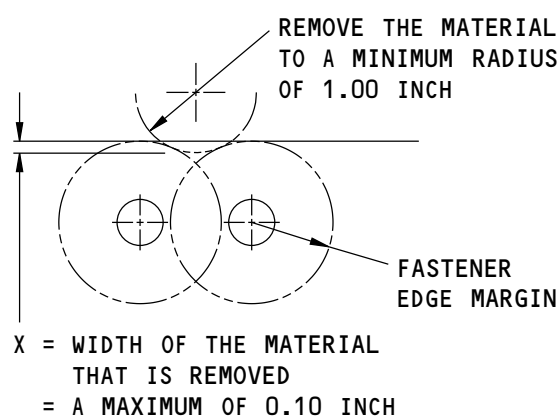
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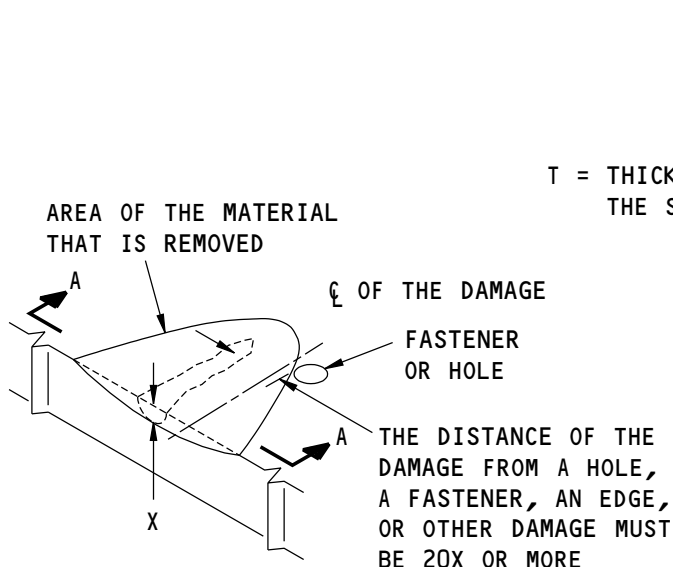
REMOVAL OF DAMAGED MATERIAL AT EDGES WHERE THE FASTENER EDGE MARGINS DO NOT HAVE AN OVERLAP

(A)



REMOVAL OF DAMAGED MATERIAL AT EDGES WHERE THE FASTENER EDGE MARGINS HAVE AN OVERLAP

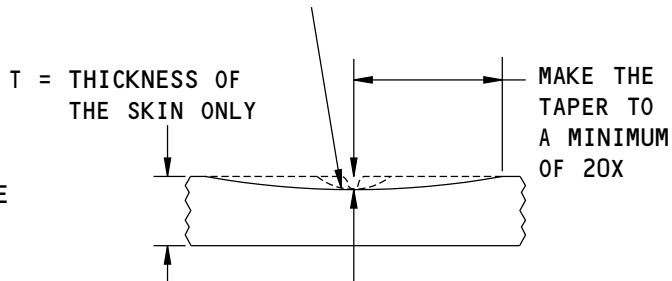
(B)



REMOVAL OF DAMAGED MATERIAL ON A SURFACE

(C)

REMOVE THE MATERIAL TO A MINIMUM RADIUS OF 1.00 INCH, THEN TAPER AS SHOWN



X = DEPTH OF THE MATERIAL THAT IS REMOVED
= A MAXIMUM OF 0.10T IN ZONE 1 UNLESS 0.05T IS SPECIFIED IN TABLE 103
= A MAXIMUM OF 0.10T TO 0.15T IN ZONES 2 AND 3 AS SHOWN IN FIGURE 104 UNLESS 0.05T IS SPECIFIED IN TABLE 103

A-A

L35660 S0006587372_V4

Allowable Damage Limits
Figure 103 (Sheet 1 of 18)

53-00-01

ALLOWABLE DAMAGE 1

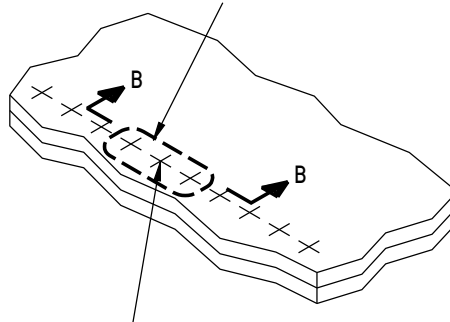
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D634A210

737-800 STRUCTURAL REPAIR MANUAL

THE REMOVAL OF MATERIAL AROUND THREE
FASTENERS IN ALL GROUPS OF TEN IS
PERMITTED TO A MAXIMUM DEPTH OF X



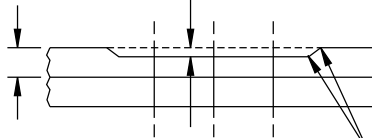
REMOVE THE FASTENERS BEFORE THE
DAMAGE IS REMOVED. INSTALL THE
FASTENERS AFTER THE REWORK IS DONE

REMOVAL OF DAMAGE AROUND THE FASTENERS ON A SURFACE

(D)

X = DEPTH OF THE MATERIAL THAT IS REMOVED
= A MAXIMUM OF 0.10T IN ZONE 1 UNLESS
0.05T IS SPECIFIED IN TABLE 103
= A MAXIMUM OF 0.10T TO 0.15T IN
ZONES 2 AND 3 AS SHOWN IN FIGURE 104
UNLESS 0.05T IS SPECIFIED IN TABLE 103

T = THICKNESS OF
THE MATERIAL



MAKE THE CONTOUR SMOOTH
TO A MINIMUM RADIUS OF
0.50 INCH (TYPICAL)

B-B

L35655 S0006587374_V3

Allowable Damage Limits Figure 103 (Sheet 2 of 18)

53-00-01

ALLOWABLE DAMAGE 1

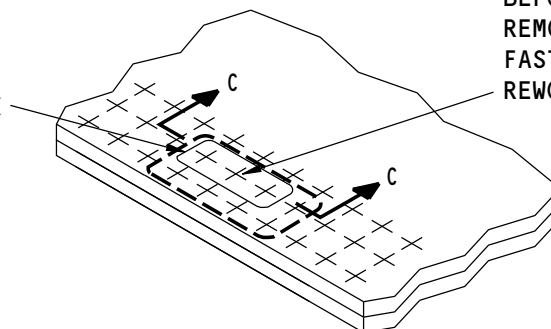
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737-800 STRUCTURAL REPAIR MANUAL

THE REMOVAL OF MATERIAL AROUND THREE FASTENERS IN ALL GROUPS OF EIGHT (3RD FASTENER ROW NOT INCLUDED) IS PERMITTED TO A MAXIMUM DEPTH OF X



REMOVE THE FASTENERS BEFORE THE DAMAGE IS REMOVED. INSTALL THE FASTENERS AFTER THE REWORK IS DONE

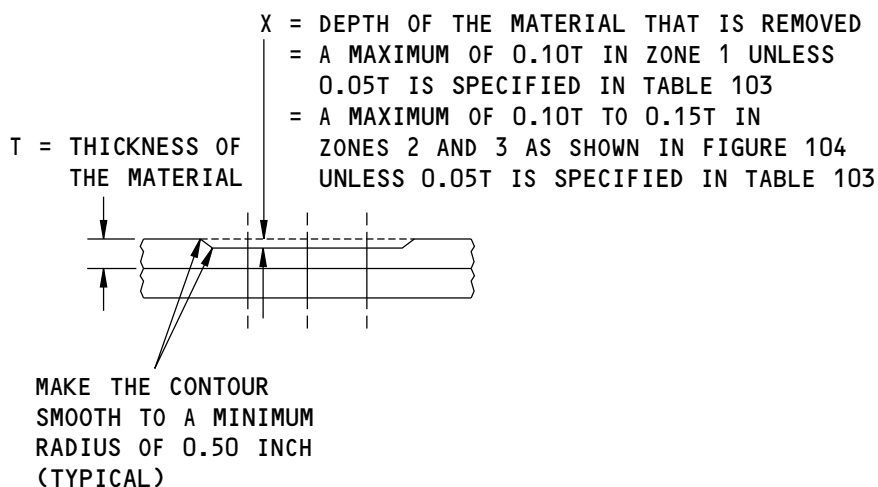
NOTE: DAMAGE REMOVAL IS SHOWN ON THE OUTER SURFACE. DAMAGE REMOVAL IS PERMITTED ON THE OTHER SURFACES AS SHOWN IN FIGURE 104.

MATERIAL REMOVAL IS NOT PERMITTED IN THE 3RD ROW FROM THE EDGE ON LAP SPLICE.

BUTT SPLICE IS SIMILAR EXCEPT: MATERIAL REMOVAL IS NOT PERMITTED IN THE 2ND ROW OF A 2-ROW BUTT. MATERIAL REMOVAL IS NOT PERMITTED IN THE 3RD ROW OF A 3-ROW BUTT.

REMOVAL OF DAMAGE AROUND THE FASTENERS OF AN EDGE OR A SURFACE AT A LAP SPLICE

(E)



NOTE: REFER TO TABLE 103

C-C

L35659 S0006587377_V3

Allowable Damage Limits Figure 103 (Sheet 3 of 18)

53-00-01

ALLOWABLE DAMAGE 1

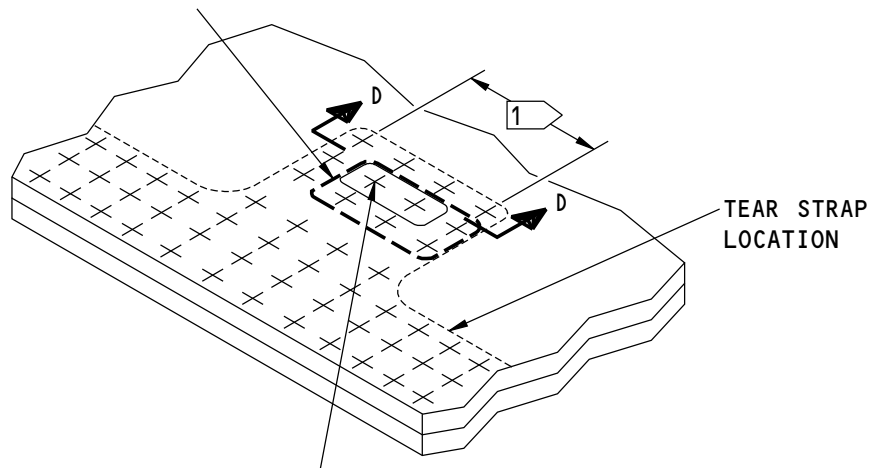
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THE REMOVAL OF MATERIAL AROUND TWO FASTENERS IN ALL GROUPS OF SIX (FIRST FASTENER ROW NOT INCLUDED) IS PERMITTED TO A MAXIMUM DEPTH OF X



REMOVE THE FASTENERS BEFORE THE DAMAGE IS REMOVED. INSTALL THE FASTENERS AFTER THE REWORK IS DONE

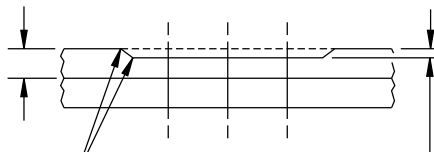
NOTE: MATERIAL REMOVAL NOT PERMITTED IN THE FIRST FASTENER ROW FROM THE EDGE OF THE TEAR STRAP TAB. DAMAGE REMOVAL IS SHOWN ON THE OUTER SURFACE.

1 THERE ARE TEAR STRAPS THAT HAVE FIVE FASTENERS ACROSS INSTEAD OF FOUR.

REMOVAL OF DAMAGE AROUND THE FASTENERS AT A TEAR STRAP TAB

(F)

T = THICKNESS OF
THE MATERIAL



MAKE THE CONTOUR SMOOTH TO A MINIMUM RADIUS OF 0.50 INCH (TYPICAL)

X = DEPTH OF THE MATERIAL THAT IS REMOVED
= A MAXIMUM OF 0.10T IN ZONE 1 UNLESS
0.05T IS SPECIFIED IN TABLE 103

D-D

L35663 S0006587380_V3

**Allowable Damage Limits
Figure 103 (Sheet 4 of 18)**

53-00-01

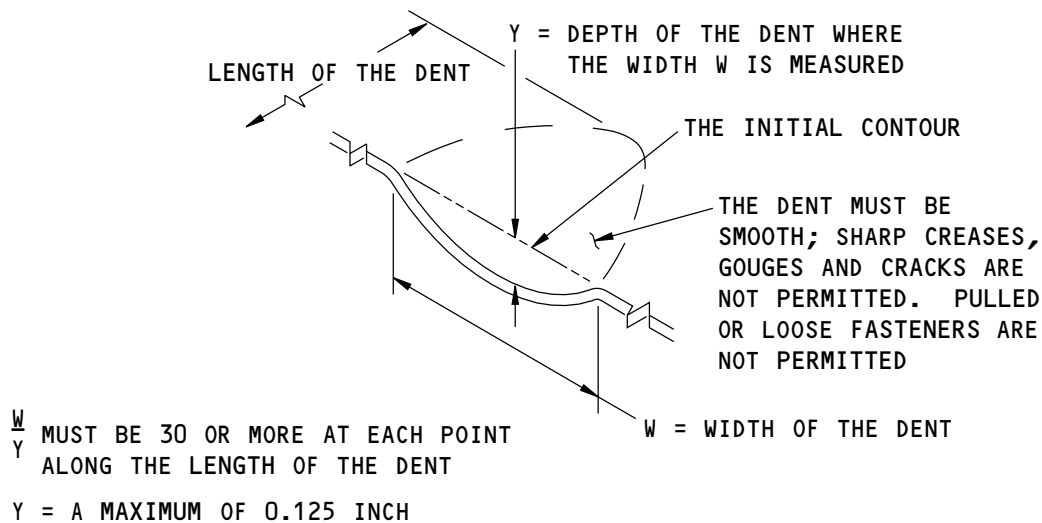
ALLOWABLE DAMAGE 1

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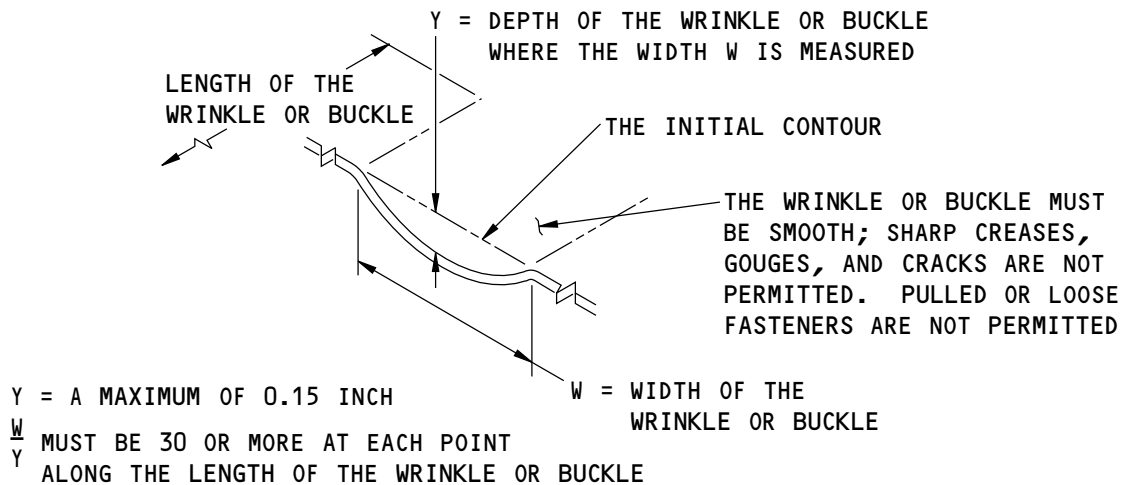
D634A210

737-800 STRUCTURAL REPAIR MANUAL



DENT THAT IS PERMITTED

(G)



WRINKLES OR BUCKLES THAT ARE PERMITTED

(H)

F53469 S0006587382_V3

Allowable Damage Limits
Figure 103 (Sheet 5 of 18)

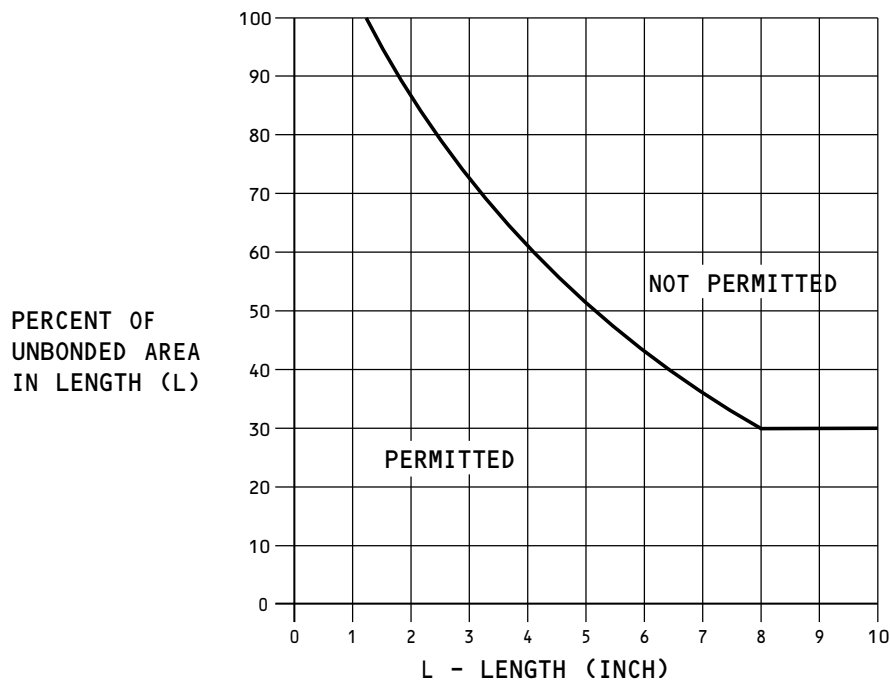
53-00-01

ALLOWABLE DAMAGE 1
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STRUCTURAL REPAIR MANUAL



NOTE: REFER TO DETAILS J AND K FOR THE DEFINITIONS OF THE LENGTH (L) AND THE UNBONDED AREA.

UNBONDED AREAS IN BONDED DOUBLERS



F53495 S0006587383_V3

Allowable Damage Limits
Figure 103 (Sheet 6 of 18)

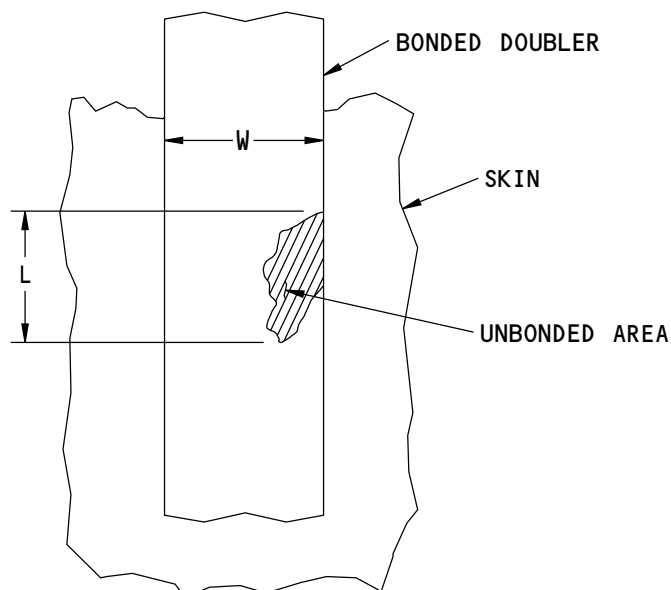
53-00-01

ALLOWABLE DAMAGE 1
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$$\text{PERCENT OF UNBONDED AREA IN LENGTH (L)} = \frac{\text{UNBONDED AREA}}{W \times L} \times 100$$

ONE UNBONDED AREA

J

G42109 S0006587384_V1

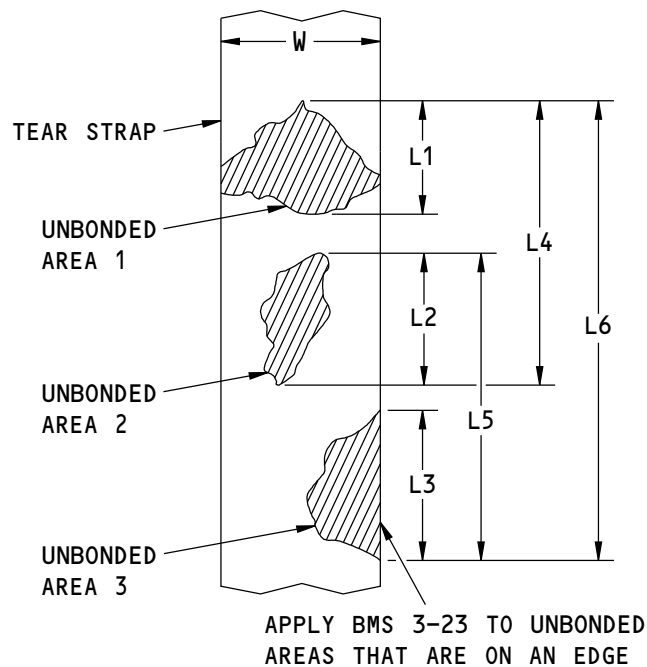
Allowable Damage Limits
Figure 103 (Sheet 7 of 18)

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ALLOWABLE DAMAGE 1
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$$\begin{aligned}
 \text{PERCENT OF UNBONDED AREA IN LENGTH L} &= \frac{\text{AREA 1}}{W \times L_1} \times 100 \\
 &= \frac{\text{AREA 2}}{W \times L_2} \times 100 \\
 &= \frac{\text{AREA 3}}{W \times L_3} \times 100 \\
 &= \frac{\text{AREA 1} + \text{AREA 2}}{W \times L_4} \times 100 \\
 &= \frac{\text{AREA 2} + \text{AREA 3}}{W \times L_5} \times 100 \\
 &= \frac{\text{AREA 1} + \text{AREA 2} + \text{AREA 3}}{W \times L_6} \times 100
 \end{aligned}$$

THE PERCENT OF UNBONDED AREA IN ANY LENGTH (L_1 THROUGH L_6 IN THIS EXAMPLE) SHOULD BE BELOW THE CURVE.

MORE THAN ONE UNBONDED AREA

(K)

G42106 S0006587385_V1

Allowable Damage Limits
Figure 103 (Sheet 8 of 18)

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ALLOWABLE DAMAGE 1
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DAMAGE DESCRIPTION	LIGHTNING STRIKE DAMAGE RATING (LSDR)
EXPOSED COUNTERSINK (WHERE PART OF THE RIVET HEAD MATERIAL IS MISSING) OR MISSING FASTENER. FOR A MISSING FASTENER BEFORE ANY REVENUE FLIGHTS ACCOMPLISH SRM 53-00-01, REPAIR 13.	15
IF THERE ARE MORE THAN FIVE (5) CONSECUTIVE DAMAGED FASTENERS IN A ROW, • APPLIES TO LAP JOINT CRITICAL ROW, BUTT JOINT, SKIN/STRINGER ATTACHMENT	15
IF THERE ARE BETWEEN 3 AND 5 CONSECUTIVE DAMAGED FASTENERS, • APPLIES TO LAP JOINT CRITICAL ROW, BUTT JOINT, SKIN/STRINGER ATTACHMENT IF THERE ARE MORE THAN EIGHT (8) CONSECUTIVE DAMAGED FASTENERS, • APPLIES TO LAP JOINT NON-CRITICAL ROW	5
IF THERE ARE 1 TO 2 CONSECUTIVE DAMAGED FASTENERS, • APPLIES TO LAP JOINT CRITICAL ROW, BUTT JOINT, SKIN/STRINGER ATTACHMENT IF THERE ARE 1 TO 8 CONSECUTIVE DAMAGED FASTENERS, • APPLIES TO LAP JOINT NON-CRITICAL ROW	2

LIGHTNING STRIKE DAMAGE RATING (LSDR)
(SEE EXAMPLE IN DETAILS N, O AND P)

L

DETAIL LEVEL	MAXIMUM LSDR FOR 10 FLIGHT CYCLE REPAIR LIMIT	MAXIMUM LSDR FOR 350 FLIGHT CYCLE REPAIR LIMIT
SINGLE SKIN PANEL	10 OR MORE	9
AIRPLANE SECTION (SUCH AS 41 SECTION)	15 OR MORE	14

AIRPLANE ALLOWABLE DAMAGE LIMIT FOR LSDR
(LIGHTNING STRIKE DAMAGE RATING)
(SEE EXAMPLE IN DETAILS N, O AND P)

M

J74262 S0000178096_V3

Allowable Damage Limits
Figure 103 (Sheet 9 of 18)

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ALLOWABLE DAMAGE 1

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STRUCTURAL REPAIR MANUAL**

DAMAGE	SCORE (USE FROM DETAIL L)	REPAIR INTERVAL (USE FROM DETAIL M)
FOR: • LAP JOINT NON-CRITICAL ROW, 11 FASTENERS • BUTT JOINT, 2 FASTENERS	7	REPAIR IN 350 FLIGHT CYCLES OR LESS
FOR: • SKIN/STRINGER, 2 FASTENERS • LAP JOINT CRITICAL ROW, 5 FASTENERS • LAP JOINT NON-CRITICAL ROW, 4 FASTENERS	9	REPAIR IN 350 FLIGHT CYCLES OR LESS
FOR 1 MISSING FASTENER	15	BEFORE ANY REVENUE FLIGHTS ACCOMPLISH SRM 53-00-01, REPAIR 13
FOR: • LAP JOINT NON-CRITICAL ROW, 5 FASTENERS • LAP JOINT CRITICAL ROW, 3 FASTENERS • SKIN/STRINGER, 8 FASTENERS	22	REPAIR IN 10 FLIGHT CYCLES OR LESS

**PANEL SECTION DAMAGE RATING
(SAMPLE CALCULATION)**

N

J74327 S0000178097_V2

**Allowable Damage Limits
Figure 103 (Sheet 10 of 18)**

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ALLOWABLE DAMAGE 1

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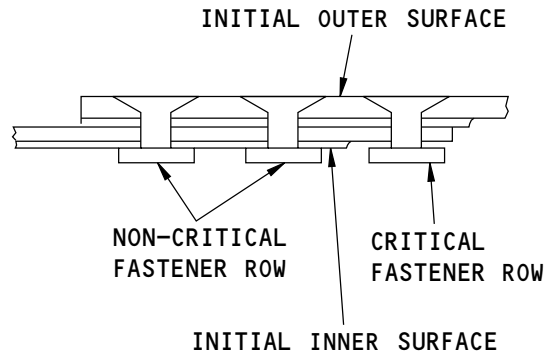
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STRUCTURAL REPAIR MANUAL

DAMAGE	SCORE (USE FROM DETAIL L)	REPAIR INTERVAL (USE FROM DETAIL M)
FOR SECTION 41 - 2 PANELS • LAP JOINT NON-CRITICAL ROW, 7 FASTENERS • BUTT JOINT, 4 FASTENERS	7	REPAIR IN 350 FLIGHT CYCLES OR LESS
FOR SECTION 48, - 3 PANELS • SKIN/STRINGER, 5 FASTENERS • LAP JOINT CRITICAL ROW, 4 FASTENERS • LAP JOINT NON-CRITICAL ROW, 4 FASTENERS	12	REPAIR IN 350 FLIGHT CYCLES OR LESS
FOR SECTION 41 - 1 PANEL • 1 MISSING FASTENER	15	BEFORE ANY REVENUE FLIGHTS ACCOMPLISH SRM 53-00-01, REPAIR 13

MAJOR SECTION DAMAGE RATING
(SAMPLE CALCULATION)



DEFINITION OF CRITICAL FASTENER ROW



J75449 S0000178561_V2

Allowable Damage Limits
Figure 103 (Sheet 11 of 18)

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ALLOWABLE DAMAGE 1

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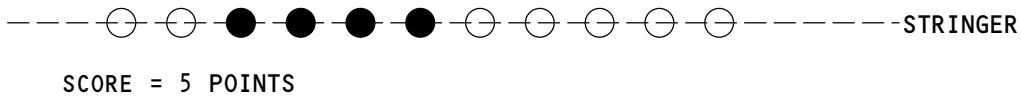
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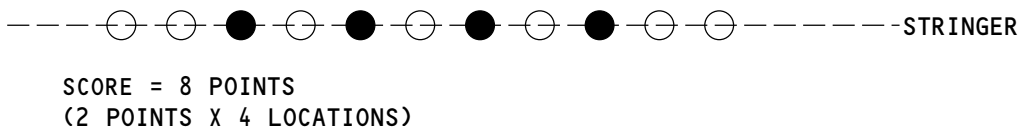
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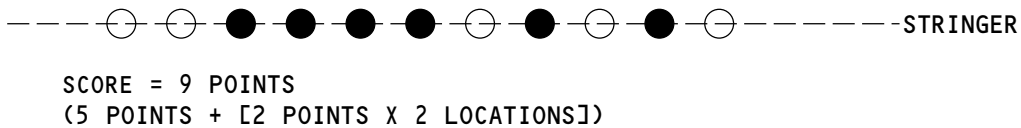
EXAMPLE A



EXAMPLE B



EXAMPLE C



KEY:

-  FASTENER (NO DAMAGE)
-  DAMAGED FASTENER

SKIN/STRINGER FASTENER DAMAGE RATING
(SAMPLE CALCULATIONS)

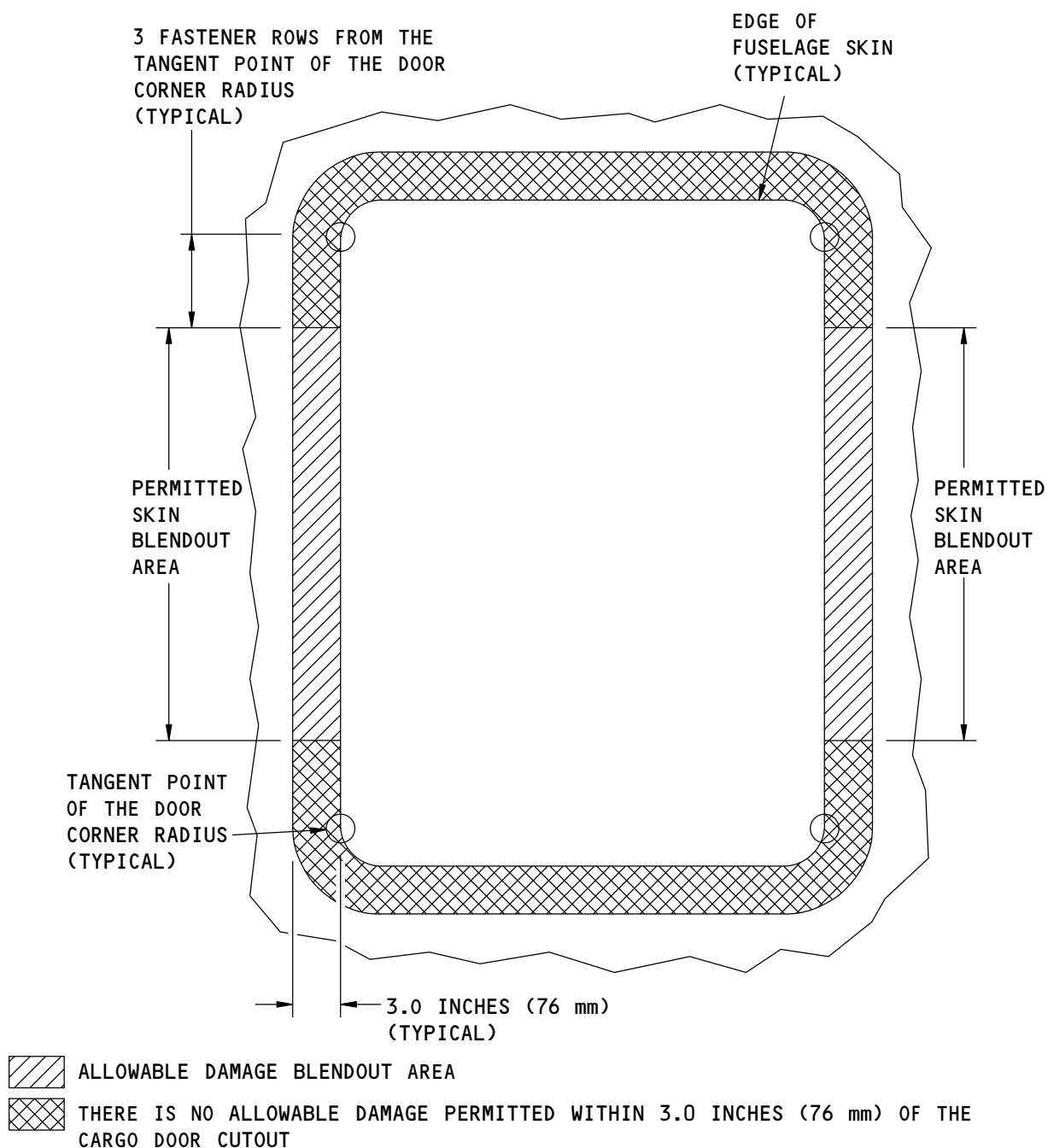
(P)

2539056 S0000602720_V1

Allowable Damage Limits
Figure 103 (Sheet 12 of 18)

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ALLOWABLE DAMAGE 1
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FORWARD AND AFT CARGO DOOR CUTOUT BLENDOUT ZONES

Q

480143 S0000143190_V7

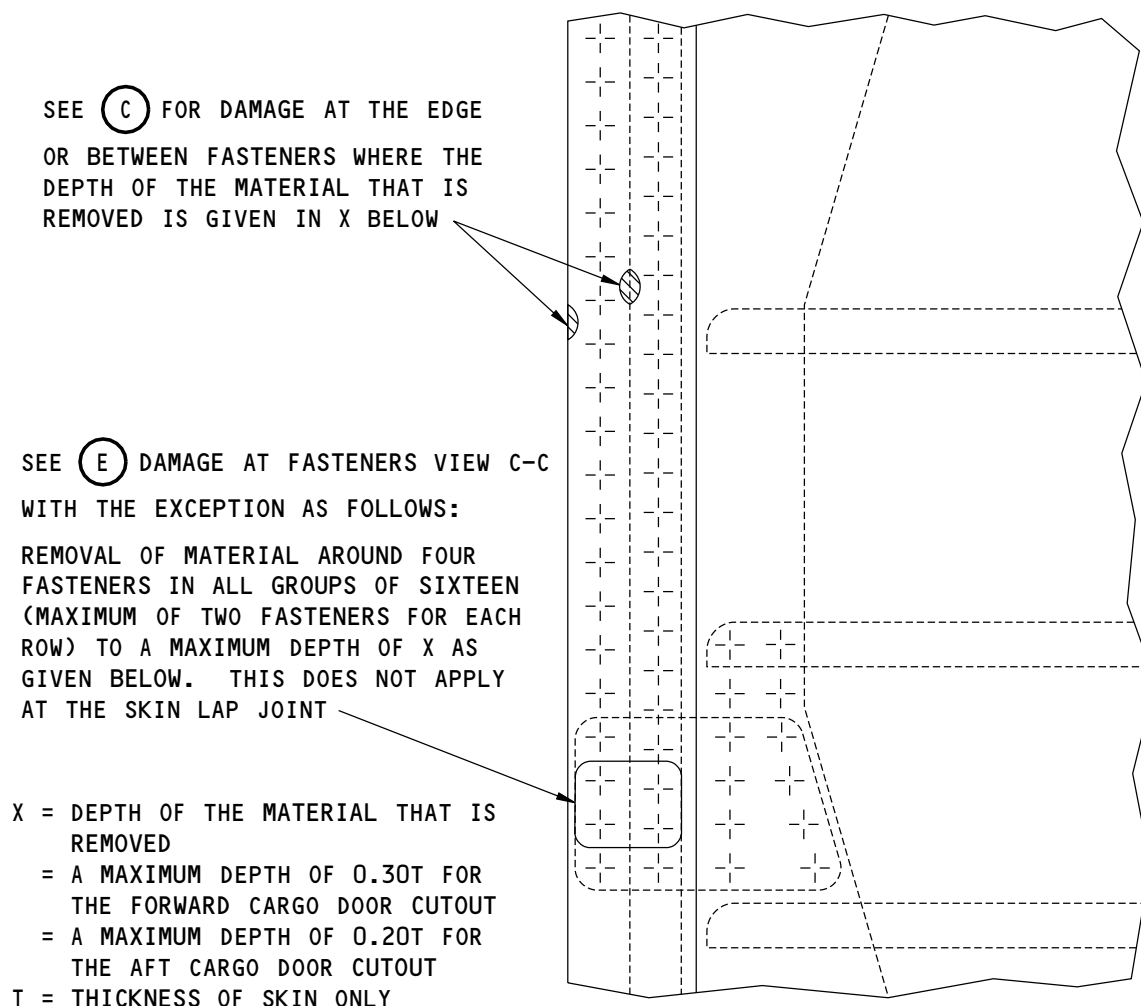
Allowable Damage Limits
Figure 103 (Sheet 13 of 18)

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ALLOWABLE DAMAGE 1
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STRUCTURAL REPAIR MANUAL**



FASTENER SYMBOLS

 REFERENCE FASTENER LOCATION.

FUSELAGE SKIN ADJACENT TO THE FORWARD AND AFT CARGO DOOR CUTOUT

(R)

480148 S0000143191_V5

**Allowable Damage Limits
Figure 103 (Sheet 14 of 18)**

53-00-01

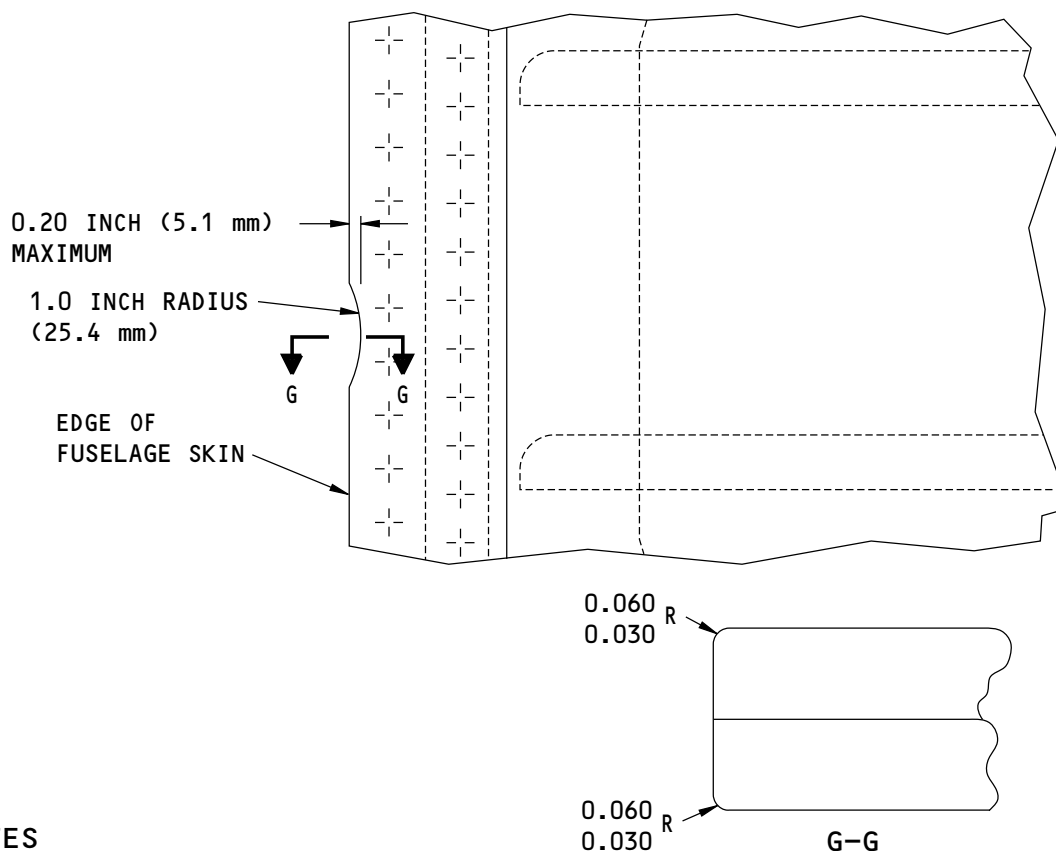
ALLOWABLE DAMAGE 1

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737-800 STRUCTURAL REPAIR MANUAL



NOTES

- TRIM OUT THE DAMAGE AS SHOWN:
 - PERFORM A HIGH FREQUENCY EDDY CURRENT (HFEC) INSPECTION ALONG THE EDGE OF THE SKIN AND/OR BEARSTRAP TRIM.
 - IF NO DAMAGE IS FOUND PERFORM AN 0.04 INCH INSURANCE CUT ALONG THE EDGE OF THE TRIM.
 - THE EDGE FINISH SHOULD BE 63Ra OR BETTER. MAINTAIN THE DRAWING EDGE CONFIGURATION AS SHOWN IN G-G.
- THE SKIN, BEAR STRAP, OR BOTH MAY BE BLENDED OUT TO A MAXIMUM DEPTH OF 0.20 INCH (5.1 mm) USING A 1.0 INCH (25.4 mm) RADIUS, AS SHOWN.
- MAKE SURE THERE IS A MINIMUM EDGE MARGIN OF 2D (WHERE D = FASTENER DIAMETER) AT ALL LOCATIONS.

FASTENER SYMBOLS

—|— REFERENCE FASTENER LOCATION.

FUSELAGE SKIN ADJACENT TO THE FORWARD AND AFT CARGO DOOR CUTOUT



480149 S0000143192_V4

Allowable Damage Limits
Figure 103 (Sheet 15 of 18)

53-00-01

ALLOWABLE DAMAGE 1

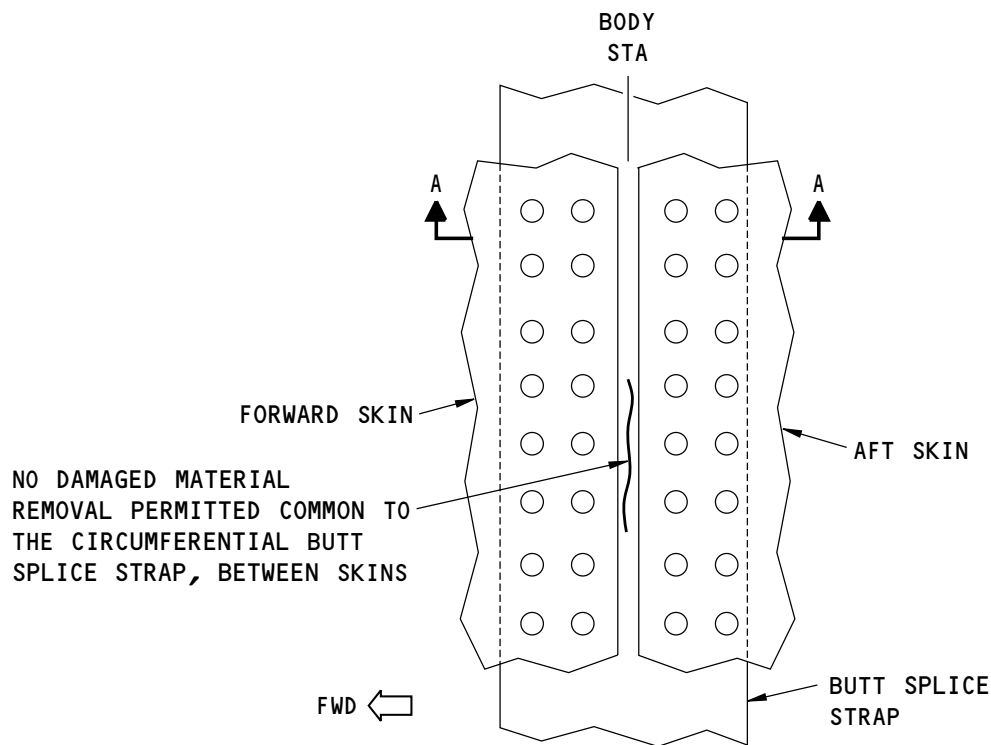
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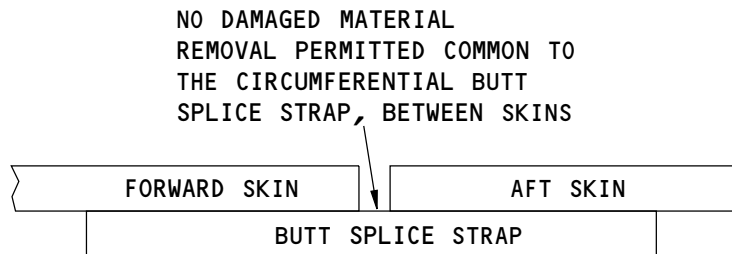
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TYPICAL SKIN BUTT JOINT



A-A

CIRCUMFERENTIAL BUTT SPLICE LIMITS



U74175 S0000215077_V2

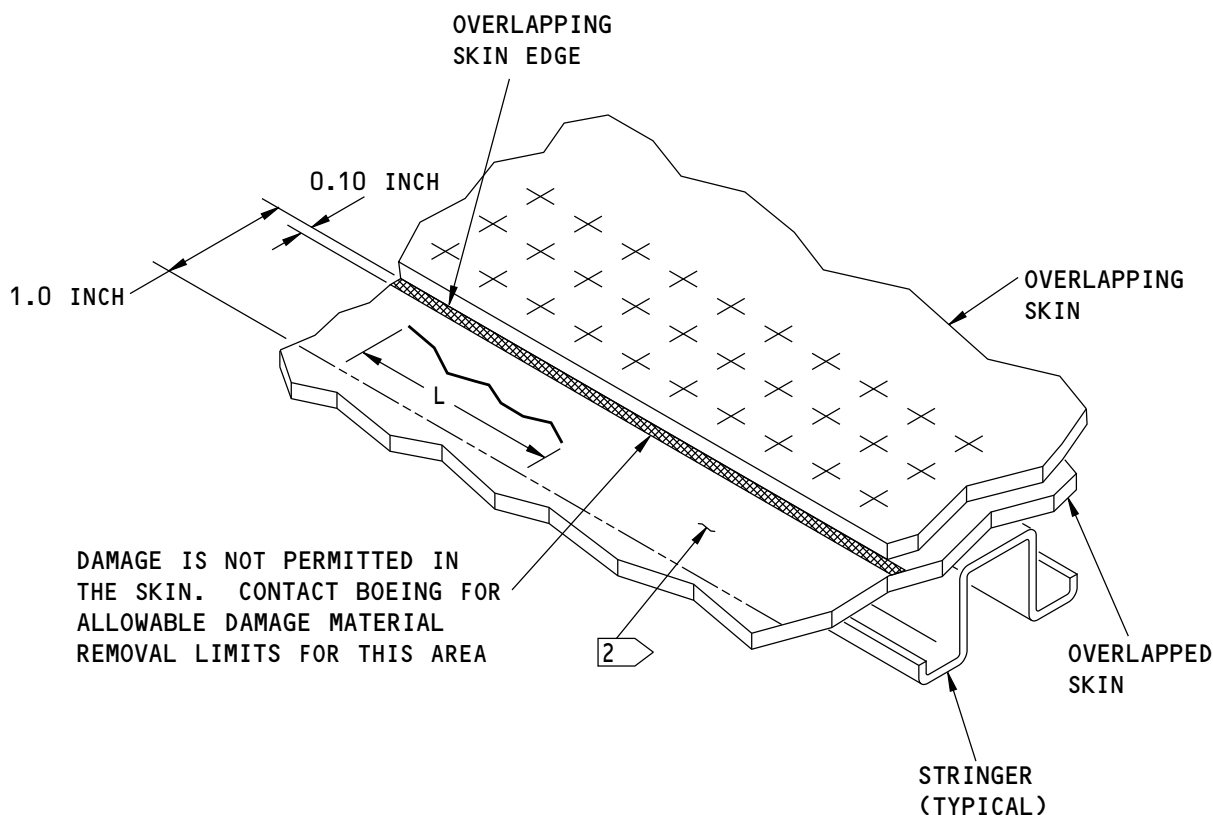
Allowable Damage Limits
Figure 103 (Sheet 16 of 18)

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ALLOWABLE DAMAGE 1
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NOTES

- 2 BETWEEN 0.10 AND 1.0 INCHES FROM OVERLAPPING SKIN EDGE, DAMAGE UP TO 2.0 INCHES IN LENGTH (L) CAN BE REMOVED AS SHOWN IN DETAIL C. DAMAGED MATERIAL REMOVAL BOUNDARY CAN EXTEND INTO THE 0.0 TO 0.10 INCHES ZONE FROM OVERLAPPING SKIN EDGE. CONTACT BOEING IF THE PROPER 20:1 BLEND RATIO CANNOT BE ACHIEVED. DAMAGED MATERIAL REMOVAL MUST BE A MINIMUM OF 40 INCHES FROM THE NEAREST ADJACENT DAMAGED MATERIAL REMOVAL OR REPAIR ALONG THE SAME LAP SPLICE. IF FILLET SEALING IS ADDED AFTER THE REMOVAL OF DAMAGED MATERIAL, THE FILLET SEAL MUST NOT COVER THE AREA OF THE ORIGINAL DAMAGE.

NOTE: KEEP ALL OTHER LIMITS AND CRITERIA AS GIVEN IN THIS ALLOWABLE DAMAGE SRM 53-00-01.

SKIN LAP JOINT



U74174 S0000215079_V2

**Allowable Damage Limits
Figure 103 (Sheet 17 of 18)**

53-00-01

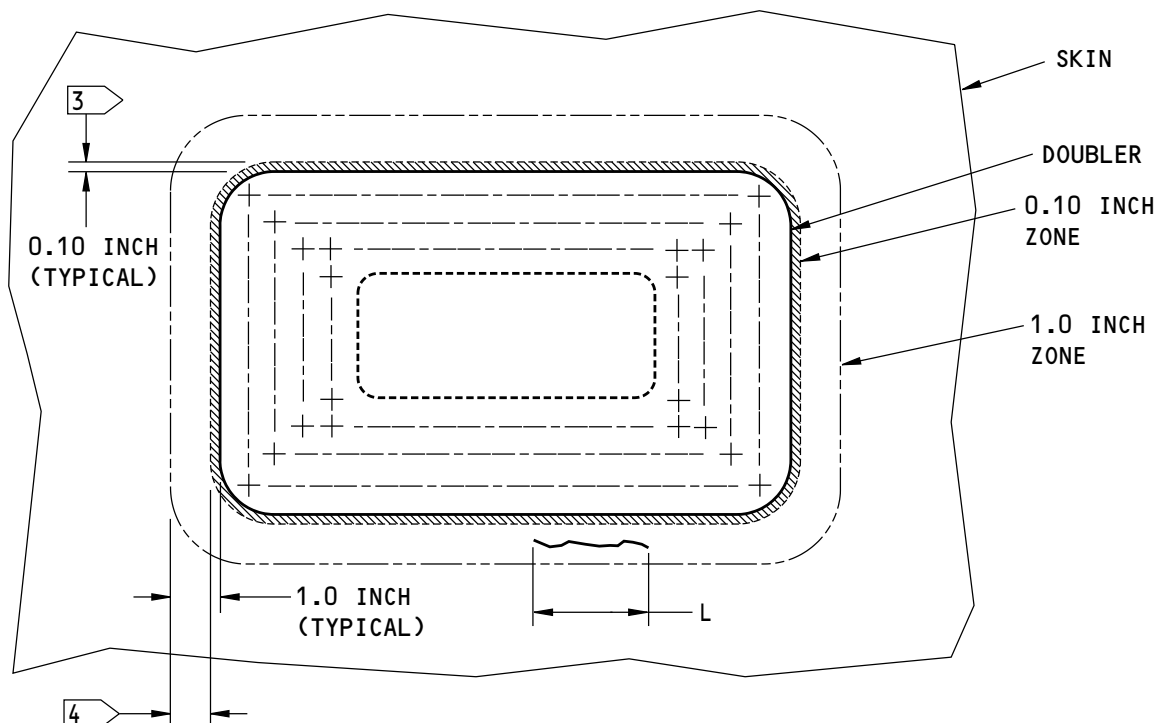
ALLOWABLE DAMAGE 1

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NOTES

3 WITHIN 0.10 INCH FROM THE EDGE OF DOUBLER, NO DAMAGE IS PERMITTED IN THE SKIN. CONTACT BOEING FOR ALLOWABLE DAMAGE MATERIAL REMOVAL LIMITS FOR THIS AREA.

4 BETWEEN 0.10 AND 1.0 INCHES FROM THE EDGE OF DOUBLER, DAMAGE UP TO 2.0 INCHES IN LENGTH (L) CAN BE REMOVED AS GIVEN IN DETAIL C. DAMAGED MATERIAL REMOVAL BOUNDARY CAN EXTEND INTO THE 0.0 TO 0.10 INCHES ZONE FROM THE DOUBLER EDGE. REMOVE DOUBLER IF NECESSARY TO ACHIEVE TAPER RATIO AS GIVEN IN DETAIL C. REINSTALL DOUBLER AND FASTENERS AS GIVEN IN APPROPRIATE SRM FIGURE OR CONTACT BOEING FOR ASSISTANCE. DAMAGED MATERIAL REMOVAL MUST BE A MINIMUM OF 40 INCHES FROM THE NEAREST AREA OF DAMAGED MATERIAL REMOVAL ADJACENT TO THE SAME DOUBLER. IF FILLET SEALING IS ADDED AFTER REMOVAL OF DAMAGED MATERIAL, THE FILLET SEAL MUST NOT COVER THE AREA OF THE ORIGINAL DAMAGE.

NOTE: KEEP ALL OTHER LIMITS AND CRITERIA AS GIVEN IN THIS ALLOWABLE DAMAGE SRM 53-00-01.

SKIN EXTERNAL DOUBLER



U74365 S0000215080_V2

**Allowable Damage Limits
Figure 103 (Sheet 18 of 18)**

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ALLOWABLE DAMAGE 1

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5. Airplane Operation Limits

A. If there is fuselage skin damage, airplane operation limits can be necessary.

- (1) Find the applicable area in Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1 for the length and depth of the damage in all 20-inch by 20-inch square areas of the fuselage skin.
 - (a) The depth of the damage shown in Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1 is given as a percentage of the initial skin thickness.
 - 1) When you calculate the damage depth, use the skin thickness given in the applicable engineering drawings. Do not include doublers. Include only the single layer of a bonded skin assembly from which material is removed.
 - (b) Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1, Detail A is applicable to damage in fuselage skin allowable damage Zone 1. Refer to Allowable Damage Zones, Figure 102/ALLOWABLE DAMAGE 1 for the definition of the fuselage skin zones.
 - (c) Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1, Detail B is applicable to damage in fuselage skin allowable Zones 2 and 3. Refer to Allowable Damage Zones, Figure 102/ALLOWABLE DAMAGE 1 for the definition of the fuselage skin zones.
 - (d) Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1 is applicable to:
 - 1) Cracks
 - 2) Nicks, Scratches, Gouges, and Corrosion
 - 3) Holes and Punctures.
 - (e) Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1 is not applicable to:
 - 1) Dents
 - 2) Wrinkles or Buckles.
- (2) Refer to Table 104/ALLOWABLE DAMAGE 1 to find the damage treatment and permitted airplane operations for the area you found in Damage Dimensions for the Fuselage Skin, Figure 104/ALLOWABLE DAMAGE 1.

Table 104:

PERMITTED AIRPLANE OPERATIONS		
FIGURE 104 CHART AREA	DAMAGE TREATMENT	PERMITTED AIRPLANE OPERATIONS
A	Clean up the damage as given in Figure 103	There are no airplane operation limits
B	Clean up the damage as given in Figure 103	Up to 50 revenue flight hours are permitted
	Do a permanent repair	There are no airplane operation limits

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ALLOWABLE DAMAGE 1

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Table 104: (Continued)

PERMITTED AIRPLANE OPERATIONS		
FIGURE 104 CHART AREA	DAMAGE TREATMENT	PERMITTED AIRPLANE OPERATIONS
C	Drill 0.25 inch diameter stop holes at the end of the cracks. Clean up other damage as given in Figure 103	A non-revenue flight to a repair station is permitted if the applicable regulatory authority gives approval before the flight. It is recommended that the proposed repair procedure be given to Boeing. The maximum cabin pressure differential is limited to 5.0 PSIG unless the skin is repaired. Cabin pressure limits are for skin damage to the pressurized fuselage skin only.
	Do an optional time-limited repair as given in SRM 53-00-01, Repair 5	Revenue flights are permitted as given in SRM 53-00-01, Repair 5
	Do a permanent repair	There are no airplane operation limits
D	Drill 0.25 inch diameter stop holes at the end of the cracks. Clean up other damage as given in Figure 103	A non-revenue flight to a repair station is permitted if the applicable regulatory authority gives approval before the flight. It is recommended that the proposed repair procedure be given to Boeing. The maximum cabin pressure differential is limited to 5.0 PSIG unless the skin is repaired. Cabin pressure limits are for skin damage to the pressurized fuselage skin only.
	Do a permanent repair	There are no airplane operation limits
E	Drill 0.25 inch diameter stop holes at the end of the cracks. Clean up other damage as given in Figure 103	A non-revenue flight to a repair station is permitted if the applicable regulatory authority gives approval before the flight. It is recommended that the proposed repair procedure be given to Boeing. The maximum cabin pressure differential is limited to zero PSIG. Cabin pressure limits are for skin damage to the pressurized fuselage skin only.
	Do a permanent repair	There are no airplane operation limits
F	Drill 0.25 inch diameter stop holes at the end of the cracks. Clean up other damage as given in Figure 103	Operation is not permitted before Boeing and the applicable regulatory authority give approval.
	Do a permanent repair	There are no airplane operation limits

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ALLOWABLE DAMAGE 1

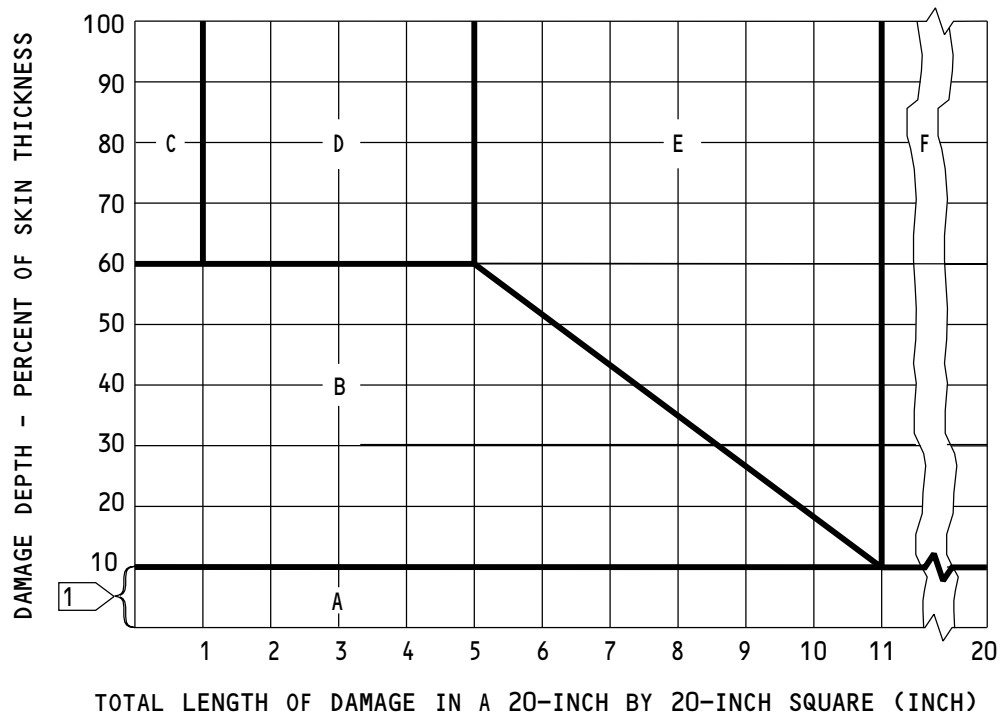
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DAMAGE DIMENSIONS FOR FUSELAGE SKIN
ALLOWABLE DAMAGE ZONE 1

A

NOTES

1 SOME LOCATIONS HAVE A 5 PERCENT DAMAGE DEPTH. REFER TO TABLE 103

L35664 S0006587387_V3

Damage Dimensions for the Fuselage Skin
Figure 104 (Sheet 1 of 2)

53-00-01

ALLOWABLE DAMAGE 1

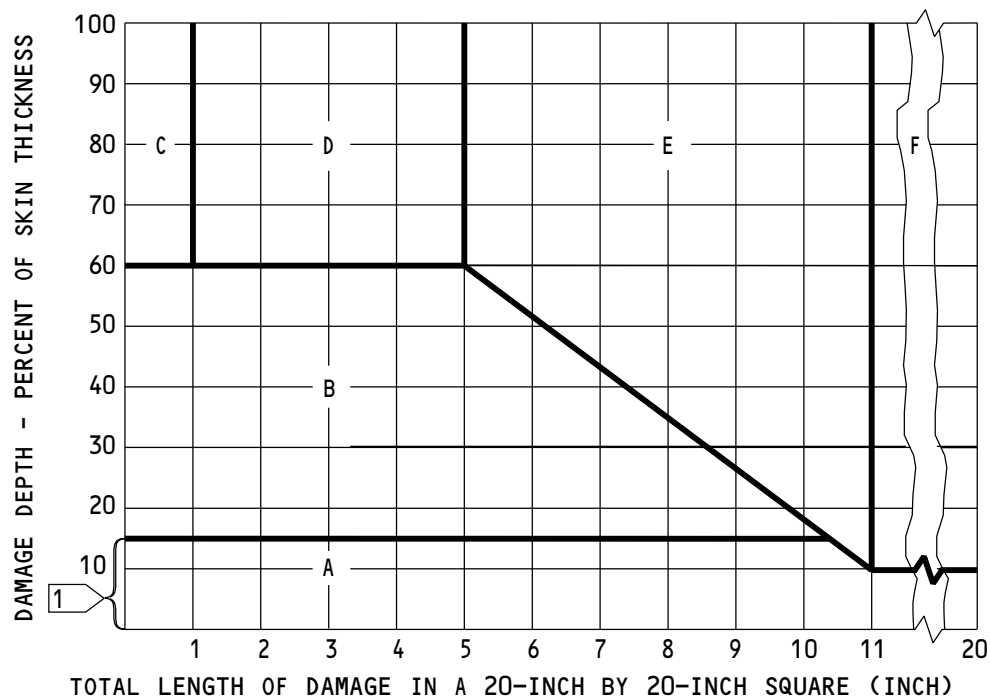
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DAMAGE DIMENSIONS FOR FUSELAGE SKIN
ALLOWABLE DAMAGE ZONES 2 AND 3

(B)

F52843 S0006587390_V1

Damage Dimensions for the Fuselage Skin
Figure 104 (Sheet 2 of 2)

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ALLOWABLE DAMAGE 1

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